

Gabriel Ryan

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Education

Ph.D., Computer Science , Columbia University (<i>advised by Prof. Suman Jana</i>)	Dec 2023
M.S., Computer Science , Columbia University	Feb 2018
B.S. Engineering & B.A. Computer Science , Swarthmore College	Jun 2013

Experience

Senior Research Scientist , Microsoft CodeAI	Mar 2024 – Present
- Training custom models for Copilot subagents, context compaction, and memory generation, enabling agentic workflows to delegate focused sub-tasks and operate over long horizons. - Led development of an efficient code embedding model used to index GitHub for Copilot's context retrieval, reducing indexing cost by 2× and index size by 3× while improving retrieval quality. - Drove increases in code survival rates through post-training improvements to the code completion models powering GitHub Copilot.	
Doctoral Researcher , Columbia Security Lab	Sep 2018 – Dec 2023
- Built a novel Linux kernel race prediction analysis using probabilistic and spectral learning. Published in Oakland S&P 2023. - Designed <i>Proximal Gradient Analysis</i> (PGA), a dataflow analysis method based on nonsmooth optimization. Implemented in LLVM. Published in USENIX Security 2021. - Designed the <i>Continuous Logic Network</i> (CLN), a neural architecture for logical learning and reasoning; applied it to learning program invariants for verification. Released a PyTorch library for constructing CLNs. Published in ICLR 2020 and PLDI 2020.	
Research Intern , AWS A.I. Labs	Jun 2023 – Aug 2023
- Built a novel LLM regression-testing approach using static analysis to prompt the model to reason symbolically about program execution paths. Achieved 2× coverage and 3× correct test generations over baseline approaches on CodeGen2, GPT-3.5, and GPT-4. - Published <i>Code-Aware Prompting</i> at FSE 2024.	
Research Intern , Microsoft Research	Jun 2021 – Aug 2021
- Built <i>TOGA: A Neural Method for Test Oracle Generation</i> using Transformers (CodeBERT) and a specialized grammar to generate unit tests, demonstrating a 170% improvement over prior work in bug detection. - Published at ICSE 2022; awarded ACM SIGSOFT Distinguished Paper Award.	
Software Engineer , Allure Security Technology	Sep 2015 – Aug 2018
- Built a web application for managing large-scale deployments of user-behavior sensors and data-loss detection using Java Spring, AngularJS, Hibernate, PostgreSQL, and MongoDB. - Designed and implemented a REST API (Swagger spec, OAuth 2.0) and a midstream network document interception and tracking system using Squid Proxy and ICAP.	
Robotics Software Consultant , 3DDataLtd	May 2015 – Aug 2015
- Built a 3D SLAM framework based on Lidar Odometry and Mapping, integrating IMU sensor data via an Extended Kalman Filter.	
Radar Software Engineer , Raytheon	Sep 2013 – Apr 2015
- Built software for radar calibration, satellite tracking, antenna diagnostics, and maintenance prioritization in Ada and C++; earned a team achievement award for delivering new diagnostic capabilities ahead of schedule.	

Selected Publications

- [ICML 2026] *SemRep: Generative Code Representation Learning with Code Transformations*. W. Li, J. Song, B. A. Stoica, A. Dhoot, G. Ryan, S. Fu, K. Pei.
- [NeurIPS DL4C Workshop 2025] *DevBench: A Realistic, Developer-Informed Benchmark for Code Generation Models*. P. A. Golnari, A. Kumarappan, W. Wen, X. Liu, G. Ryan, Y. Sun, S. Fu, et al.
- [OOPSLA 2024] *Accurate Data Race Prediction in the Linux Kernel through Sparse Fourier Learning*. G. Ryan, B. Cetin, Y. Lim, S. Jana. <https://dl.acm.org/doi/pdf/10.1145/3649840>
- [FSE 2024] *Code-Aware Prompting: A study of Coverage Guided Test Generation in Regression Setting using LLM*. G. Ryan, S. Jain, M. Shang, S. Wang, X. Ma, M. Ramanathan, B. Ray. <https://arxiv.org/pdf/2402.00097>
- [Oakland S&P 2023] *Precise Detection of Kernel Data Races with Probabilistic Lockset Analysis*. G. Ryan, A. Shah, D. She, S. Jana. https://cs.columbia.edu/~gabe/files/oakland2023_pla.pdf
- [ICSE 2022] *TOGA: A Neural Method for Test Oracle Generation*. E. Dinella*, G. Ryan*, T. Mytkowicz, S. Lahiri. <https://arxiv.org/pdf/2109.09262.pdf> (**Distinguished Paper Award**)
- [OSDI 2021] *DistAI: Data-Driven Automated Invariant Learning for Distributed Protocols*. J. Yao, R. Tao, R. Gu, J. Nieh, S. Jana, G. Ryan. <https://www.usenix.org/system/files/osdi21-yao.pdf> (**Best Paper Award**)
- [USENIX Security 2021] *Fine Grained Dataflow Tracking with Proximal Gradients*. G. Ryan, A. Shah, D. She, K. Bhat, and S. Jana. <https://arxiv.org/abs/1909.03461>
- [PLDI 2020] *Learning Nonlinear Loop Invariants with Gated Continuous Logic Networks*. J. Yao*, G. Ryan*, J. Wong*, S. Jana, and R. Gu. <https://arxiv.org/abs/2003.07959>
- [ICLR 2020] *CLN2INV: Learning Loop Invariants with Continuous Logic Networks*. G. Ryan*, J. Wong*, Y. Jianan*, R. Gu, and S. Jana. <https://arxiv.org/abs/1909.11542>

Awards

- National Defense Science and Engineering Graduate Fellowship** (NDSEG). Awarded NDSEG Fellowship for proposal "Proximal Gradient Analysis for Vulnerability Detection and Defense." 2019.
- NSF Graduate Research Fellowship Honorable Mention**. Accorded honorable mention for proposal "Modeling and Simulating Adversarial User Behavior with Sequential VAEs." 2018.

Academic Service

Workshop on Large Language Models for Code, ICSE 2026. Program Committee.

Next-Gen Code Development with Collaborative AI Agents Workshop, AAAI 2026. Program Committee.

AAAI 2024, 2025, 2026. Program Committee.

TOSEM 2023, 2024. Reviewer.

AAAI 2023. Reviewer.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, Ada.

ML & LLMs: PyTorch, Transformers, HuggingFace, LLM post-training (SFT, DPO, RLHF), prompt engineering, evaluation, data curation.

Program Analysis & Systems: LLVM, Z3, static & dynamic analysis, Linux kernel, fuzzing.

Infrastructure: Linux, Git, Docker, Azure, AWS, distributed training.